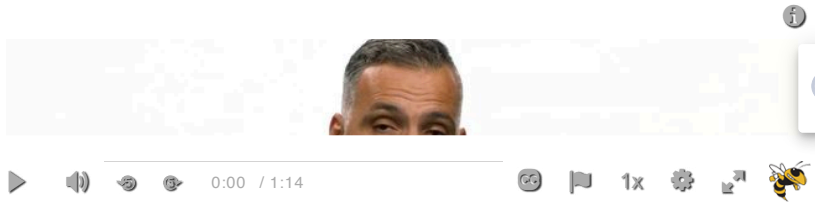


Lesson Eight Overview



Throughout the lecture, we are giving you some **“food for thought”** -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Identify overlapping communities
- Generate synthetic networks with known community structure
- Evaluate and compare community structures
- Explain the role of nodes depending on their position between/within communities
- Identify dynamic/evolving communities

Required and Recommended Reading

Required Reading

- Chapter-9 from A-L. Barabási, [Network Science](http://www.networksciencebook.com/) (<http://www.networksciencebook.com/>), 2015.

Recommended Reading

- **“Dynamic Community Detection”**  (https://doi.org/10.1007/978-1-4614-6170-8_383), by Rémy Cazabet, Frédéric Amblard, 05 October 2014.
- **“*δ*-MAPS: from spatio-temporal data to a weighted and lagged network between functional domains”** (<https://link.springer.com/article/10.1007/s41109-018-0078-z>), by Fountalis, I., Dovrolis, C., Bracco, A. et al., 2018

Assignments

☐ **L8: Knowledge Check**

L8: Quiz

Module Three: Assignment